GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Mechanical Data Sheets for Vent Stack**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-DSH-004
Document Revision	R01
Document Status	Issued For Review
Originator	Sunday Ikechukwu
Security Classification	Restricted
Issue Date	**9 Aug 2022**

Mechanical Data Sheets for Vent Stack

R01	09.08.22	Issued For Review	S.Ikechukwu	E. Nwachukwu	A. Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R01	09.08.22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-MPP-DSH-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

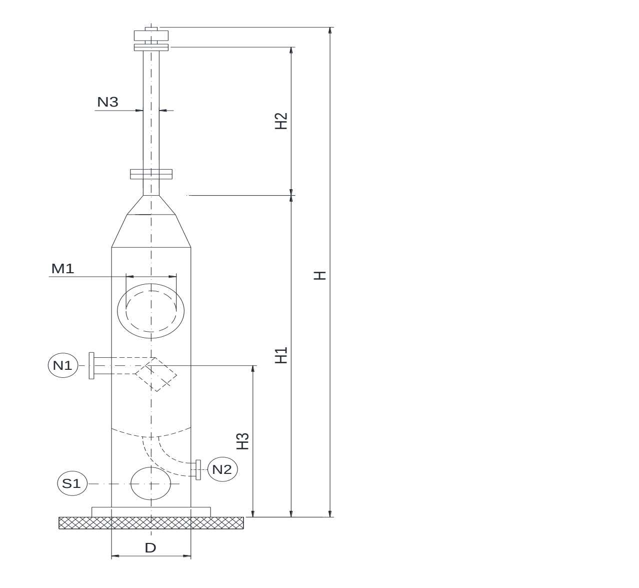
Document No.	Description

Table of Contents

	MECHANICAL DATA SHEET FOR VENT STACK - BBOU				
1	**GENERAL**				
2	Service	Orientation	Gas Venting	Vertical	
3	Site	No. Required	**BBOU MANIFOLD STATION**	One	
4	Hazardous Area Class	Gas Group	Zone 1	II B; T3	
5	Ambient temp (Min/Max)	Relative Humidity	21.4°C to 33.5°C	100 %	
6	Other Conditions		Frequent rain, Harmattan		
7	Design	Self-supporting (stand-alone)		Applicable Specifications:	
8	Design Code	ASME Boiler and Pressure Vessel Code, SectionVIII, Div 1		EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-004 SPECIFICATION FOR VENT STACK	
9	**DESIGN PARAMETERS**				
10	Medium			Natural Gas	
11	Design	Temperature		**-39/110°C**	
12	Design	Pressure		10 barg	
13	Design	Capacity		HOLD 1	
14					
15	Operating	Temperature			
16	Operating	Pressure	Stack Tip	atmospheric	
17	Operating	Pressure	Body Structure	atmospheric	
18	Operating	Capacity		43710.47 kg/h (HOLD 1)	(Note 4)
19	Hydrostatic Testing	Body of Base Structure		Per Code	
20	Hydrostatic Testing	Vent Stack Extension		Per Code	
21	Gas Velocity at Vent Stack Tip			150 m/s	(Note 2)
22	Sound Pressure Level			85 dB(A) max	(Note 2)
23	Height of the Vent Stack			15 m (HOLD 1)	(Note 3)
24	Tip Diameter			304.8 mm (12 inches) (HOLD 1)	
25	Vent Stack Riser Diameter			304.8 mm (12 inches) (HOLD 1)	
26	Maximum Sterile Area radius			HOLD1	
27	Max. back pressure at base of VentStack			0.07 barg	
28	Vent Stack Tip Type			Rain Deflector with Bird Screen	
29	**GAS Composition**				
30	Refer to EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 Basis for Design				
31	**Mechanical Design Data**				
32	Test Pressure (barg)	Internal Vacuum (mm Hg)	Per UG 99b ASME Code	***N/A***	
33	Internal Diameter of Shell (mm)	Height of Vent Stack	1016 mm (40 inches)	15000	
34	Wall Thickness Shell (mm)	Wall Thickness Head (mm)	14 (VTC)	14 (VTC)	
35	Wall Thick Stack Riser (mm)	Reducer Thickness, mm	Sch. 20	Sch. 20	
36	Corrosion Allowance (mm)	Joint Efficiency	3	1	
37	Head Form	Seismic Design	2:1 Ellipsoidal	None	
38	Vessel Orientation	Total Volume Max. Oper. (m^3)	Vertical	N/A	
39	Empty Weight (kg)	Weight Full of Water (kg)	3600 (VTC)	8000 (VTC)	
40	Internals	Explosion Blast Load (kN/m^2)	N/A	None	
41	Insulation Thickness (mm)	Design Wind Speed (m/s)	NA	40	
42	**Material Specification (Note 4)**				
43	Body Structure	Head	SA 516 Gr 70 (S5)	SA 516 Gr. 70 (S5)	
44	Vent Stack Riser/Tip	Concentric Reducer	SA 333 Gr6	SA 420 WPL6	
45	Lining / Cladding	Skirt	N/A	SA 516 Gr. 70 (S5)	
46	Structural Shapes	Internals	SA 516 Gr 70 (S5)	N/A	
47	Gaskets	Lifting Lugs	Spiral Wound 316 SS	SA 516 Gr 70 (S5)	
48	Flanges	Nozzle Necks	SA 350 LF2 Class 1	SA 333 Gr6	
49	External Bolts	External Nuts	SA 320 L7	SA 194 7 (S3)	
50	Internal Bolts	Internal Nuts	NA	NA	
51	Plates & Pads	Drip Tray	SA 516 Gr 70 (S5)	N/A	
52	Vortex Breaker	Inlet Deflector	N/A	N/A	
53	**Fabrication and Inspection Requirements**				
54	Construction in accordance with:	Post Weld Heat Treatment	ASME Sect VIII, Div 1	As per code	
55	Inspection	Radiography	Per Code Requirement	100%	
56	Inspection Authority	Material Certificates	TPI	Per EN 10204 (Note 7)	
57	Special Heat Treatment	Ultrasonic	VTA	100%	
58	Magnetic Particle	Impact Tests	100%	Yes (Note 8)	
59	Dye Penetrant	Hydrostatic Testing	100%	Per Code Requirement	
60	**Other Requirements**				
61	Anchor Bolts	PWHT Required	ASTM F1554 (Note 15)	Per Code	
62	Fireproofing Support	Nameplate	Note 12	316 SS	
63	Drip Tray	Painting	N/A	Note 17	
64	Earthing Bosses/Clips	Lifting Lugs	Note 10	Note 11	
65	Ladder, Platform, Piping Clips:	Stiffening Rings	N/A	N/A	

MECHANICAL DATA SHEET FOR VENT STACK - BBOU

66 **SKETCH**



66 **VENT STACK DIMENSIONS**

67	N1	6"	D	48"
68	N2	2"	H	15 m
69	N3	12"	H1	***8667 mm***
70	M1	24"	H2	***4833 mm***
71				

72	Skirt height, mm	***1300***	H1
73	Shell body height, mm	***7300***	H1
74	Reducer 24x12 length, mm	356	H1
75	Reducer 40x24 length, mm	711	H1
76	Vent Tip height , mm	***4833***	H2
77	Vent Nozzle Inlet elevation, mm	***7000***	H3
78			

79 **NOZZLES TABLE**

	No	Size	Rating	Service	Type/Facing	Remarks
81	N1	6"	150#	Inlet	WN/RF	
82	N2	2"	150#	Drain	WN/RF	
83	M1	24"	150#	Manhole	WN/RF	With blind flange&davit
84	S1	18"		Skirt Access		

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91 **Notes**

92 1) Vendor's scope shall include complete equipment design, material, fabrication,inspection and testing.
93 2) At the tip of the Vent Stack.
94 3) From bottom surface of the skirt's baseplate up to the tip of the vent stack extension.
95 4) At full blowdown load.
96 5) Flanges shall be delivered in accordance standard ASME B16.5, RF, WN, Class 150
97 6) Material for flange: ASTM A350-LF2.
98 7) The material test certificates shall be in accordance with EN10204 Type 3.1 for all metals welded to the pressure part.
99 8) Impact test shall be made in accordance with ASTM A350 and ASTM A333, test temperature shall be -45°C.
100 9) Vent Stack shall be connected to the station's grounding system
101 10) Two earthing clips shall be provided at the skirt, approximately 500 mm above the bottom surface of the skirt's foundation flange
102 ***11) Separate lifting and tailing lugs/trunions shall be provided for the top portion.***
103 ***12) deleted***
104 13) All nozzles with size 2"and below shall be provided with 2 stiffeners 90 deg apart welded to nozzle neck and shell.
105 14) Any Support Cleat or Pad directly welded with vessel shall be same or equivalent material as vessel.
106 15) Anchor bolt design shall be carried out by vendor and relevant documentations to aid foundation design provided.
107 16) Hydrostatic test requirements shall be as per UG 99b of ASME Section VIII, Division 1 Code.
108 17) Painting shall comply with the requirements of the Painting and Coating Specification referenced herein.
109 18.) Vent Stack extension shall be equipped with vibration damper to reduce vibration at the tip of the stack.
110 ***19) Vent stack shall be U stamped in accordance with ASME Section VIII Div.1.***
111 ***20) Requirement of lightening arrestor to prevent auto ignition during lightening and emergency depressurization (considering API 520 and NFPA 780 requirements) shall be provided by Vendor. Vendor to also review the probability of auto ignition due to static charge built up and recommend/provide preventive measure.***
112 ***21) Design of vent stack tip shall be carried out by vendor for properly dilution of gases in atmosphere at below lower flammable limits.***
113 ***22) Vendor to carry out vibration analysis and limit the wind deflection. Reducer and flaring of skirt shall be provided to limit the wind deflection by H/200m where H is overall height of stack from base.***
114 ***23) Vendor to provide saddle support for transportation of Cold Vent stack.***
115 ***24) Supply of blinds required for hydrotest shall be considered and supplied by Vendor.***
116 ***25) Vendor shall evaluate the need of a strengthening arrangement (guy wires, stiffeners etc.) due to high winds and the resultant stresses on such slender structure. If required, guy wires along with their accessories shall be supplied by the vent stack Vendor.***
117 ***26). Vent Stack shall be equipped with Spark Arrestor and Bird Screen. Vendor to provide.***

118 **References**

119 1. EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 Basis for Design
120 2. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-004 Specification for Vent Stack
121 3. EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 Process Flow Sheets 1 to 8
122 4. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005 Piping Class Specification

123 **Abbreviations**

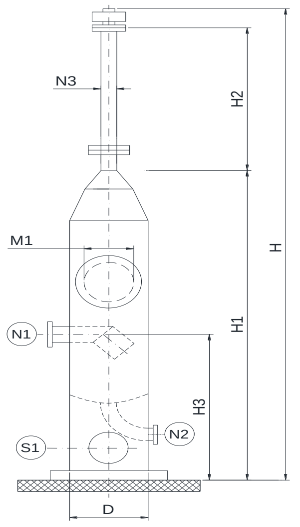
124 VTA - Vendor To Advise, VTC, Vendor to Confirm, N/A - Not Applicable

MECHANICAL DATA SHEET FOR VENT STACK - ELEPA

1	**GENERAL**				
2	Service	Orientation	Gas Venting	Vertical	
3	Site	No. Required	**ELEP MANIFOLD STATION**	One	
4	Hazardous Area Class	Gas Group	Zone 1	II B; T3	
5	Ambient temp (Min/Max)	Relative Humidity	21.4°C to 33.5°C	100 %	
6	Other Conditions		Frequent rain, Harmattan		
7	Design	Self-supporting (stand-alone)	Applicable Specifications:		
8	Design Code	ASME Boiler and Pressure Vessel Code, SectionVIII, Div 1	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-004 SPECIFICATION FOR VENT STACK		
9	**DESIGN PARAMETERS**				
10	Medium		Natural Gas		
11	Design	Temperature		**-39/110°C**	
12		Pressure		10 barg	
13		Capacity		HOLD 1	
14					
15	Operating	Temperature			
16		Pressure	Stack Tip	atmospheric	
17			Body Structure	atmospheric	
18		Capacity		43710.47 kg/h (HOLD 1)	(Note 4)
19	Hydrostatic Testing	Body of Base Structure		Per Code	
20		Vent Stack Extension		Per Code	
21	Gas Velocity at Vent Stack Tip			150 m/s	(Note 2)
22	Sound Pressure Level			85 dB(A) max	(Note 2)
23	Height of the Vent Stack			15 m (HOLD 1)	(Note 3)
24	Tip Diameter			304.8 mm (12 inches) (HOLD 1)	
25	Vent Stack Riser Diameter			304.8 mm (12 inches) (HOLD 1)	
26	Maximum Sterile Area radius			HOLD1	
27	Max. back pressure at base of VentStack			0.07 barg	
28	Vent Stack Tip Type			Rain Deflector with Bird Screen	
29	**GAS Composition**				
30	Refer to EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 Basis for Design				
31	**Mechanical Design Data**				
32	Test Pressure (barg)	Internal Vacuum (mm Hg)	Per UG 99b ASME Code	***N/A***	
33	Internal Diameter of Shell (mm)	Height of Vent Stack	1016 mm (40 inches)	15000	
34	Wall Thickness Shell (mm)	Wall Thickness Head (mm)	14 (VTC)	14 (VTC)	
35	Wall Thick Stack Riser (mm)	Reducer Thickness, mm	Sch. 20	Sch. 20	
36	Corrosion Allowance (mm)	Joint Efficiency	3	1	
37	Head Form	Seismic Design	2:1 Ellipsoidal	None	
38	Vessel Orientation	Total Volume Max. Oper. (m^3)	Vertical	N/A	
39	Empty Weight (kg)	Weight Full of Water (kg)	3600 (VTC)	8000 (VTC)	
40	Internals	Explosion Blast Load (kN/m^2)	N/A	None	
41	Insulation Thickness (mm)	Design Wind Speed (m/s)	NA	40	
42	**Material Specification (Note 4)**				
43	Body Structure	Head	SA 516 Gr 70 (S5)	SA 516 Gr. 70 (S5)	
44	Vent Stack Riser/Tip	Concentric Reducer	SA 333 Gr6	SA 420 WPL6	
45	Lining / Cladding	Skirt	N/A	SA 516 Gr. 70 (S5)	
46	Structural Shapes	Internals	SA 516 Gr 70 (S5)	N/A	
47	Gaskets	Lifting Lugs	Spiral Wound 316 SS	SA 516 Gr 70 (S5)	
48	Flanges	Nozzle Necks	SA 350 LF2 Class 1	SA 333 Gr6	
49	External Bolts	External Nuts	SA 320 L7	SA 194 7 (S3)	
50	Internal Bolts	Internal Nuts	NA	NA	
51	Plates & Pads	Drip Tray	SA 516 Gr 70 (S5)	N/A	
52	Vortex Breaker	Inlet Deflector	N/A	N/A	
53	**Fabrication and Inspection Requirements**				
54	Construction in accordance with:	Post Weld Heat Treatment	ASME Sect VIII, Div 1	As per code	
55	Inspection	Radiography	Per Code Requirement	100%	
56	Inspection Authority	Material Certificates	TPI	Per EN 10204 (Note 7)	
57	Special Heat Treatment	Ultrasonic	VTA	100%	
58	Magnetic Particle	Impact Tests	100%	Yes (Note 8)	
59	Dye Penetrant	Hydrostatic Testing	100%	Per Code Requirement	
60	**Other Requirements**				
61	Anchor Bolts	PWHT Required	ASTM F1554 (Note 15)	Per Code	
62	Fireproofing Support	Nameplate	Note 12	316 SS	
63	Drip Tray	Painting	N/A	Note 17	
64	Earthing Bosses/Clips	Lifting Lugs	Note 10	Note 11	
65	Ladder, Platform, Piping Clips:	Stiffening Rings	N/A	N/A	

MECHANICAL DATA SHEET FOR VENT STACK - ELEPA

SKETCH



VENT STACK DIMENSIONS

N1	6"	D	48"
N2	2"	H	15 m
N3	12"	H1	***8667 mm***
M1	24"	H2	***4833 mm***

Skirt height, mm	***1300***	H1
Shell body height, mm	***7300***	H1
Reducer 24x12 length, mm	356	H1
Reducer 40x24 length, mm	711	H1
Vent Tip height , mm	***4833***	H2
Vent Nozzle Inlet elevation, mm	***7000***	H3

NOZZLES TABLE

No	Size	Rating	Service	Type/Facing	Remarks
N1	6"	150#	Inlet	WN/RF	
N2	2"	150#	Drain	WN/RF	
M1	24"	150#	Manhole	WN/RF	With blind flange&davit
S1	18"		Skirt Access		

Notes

1) Vendor's scope shall include complete equipment design, material, fabrication,inspection and testing.

2) At the tip of the Vent Stack.

3) From bottom surface of the skirt's baseplate up to the tip of the vent stack extension.

4) At full blowdown load.

5) Flanges shall be delivered in accordance standard ASME B16.5, RF, WN, Class 150

6) Material for flange: ASTM A350-LF2.

7) The material test certificates shall be in accordance with EN10204 Type 3.1 for all metals welded to the pressure part.

8) Impact test shall be made in accordance with ASTM A350 and ASTM A333, test temperature shall be -45°C.

9) Vent Stack shall be connected to the station's grounding system

10) Two earthing clips shall be provided at the skirt, approximately 500 mm above the bottom surface of the skirt's foundation flange

11) Separate lifting and tailing lugs/trunions shall be provided for the top portion.

12) deleted

13) All nozzles with size 2"and below shall be provided with 2 stiffeners 90 deg apart welded to nozzle neck and shell.

14) Any Support Cleat or Pad directly welded with vessel shall be same or equivalent material as vessel.

15) Anchor bolt design shall be carried out by vendor and relevant documentations to aid foundation design provided.

16) Hydrostatic test requirements shall be as per UG 99b of ASME Section VIII, Division 1 Code.

17) Painting shall comply with the requirements of the Painting and Coating Specification referenced herein.

18.) Vent Stack extension shall be equipped with vibration damper to reduce vibration at the tip of the stack.

19) Vent stack shall be U stamped in accordance with ASME Section VIII Div.1.

20) Requirement of lightening arrestor to prevent auto ignition during lightening and emergency depressurization (considering API 520 and NFPA 780 requirements) shall be provided by Vendor. Vendor to also review the probability of auto ignition due to static charge built up and recommend/provide preventive measure.

21) Design of vent stack tip shall be carried out by vendor for properly dilution of gases in atmosphere at below lower flammable limits.

22) Vendor to carry out vibration analysis and limit the wind deflection. Reducer and flaring of skirt shall be provided to limit the wind deflection by H/200m where H is overall height of stack from base.

23) Vendor to provide saddle support for transportation of Cold Vent stack.

24) Supply of blinds required for hydrotest shall be considered and supplied by Vendor.

25) Vendor shall evaluate the need of a strengthening arrangement (guy wires, stiffeners etc.) due to high winds and the resultant stresses on such slender structure. If required, guy wires along with their accessories shall be supplied by the vent stack Vendor.

26). Vent Stack shall be equipped with Spark Arrestor and Bird Screen. Vendor to provide.

References

1. EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 Basis for Design
2. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-004 Specification for Vent Stack
3. EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 Process Flow Sheets 1 to 8
4. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005 Piping Class Specification

Abbreviations

VTA - Vendor To Advise, VTC, Vendor to Confirm, N/A - Not Applicable

	MECHANICAL DATA SHEET FOR VENT STACK-AKABEL				
1	**GENERAL**				
2	Service	Orientation	Gas Venting	Vertical	
3	Site	No. Required	**AKABEL PIG RECEIVER STATION**	One	
4	Hazardous Area Class	Gas Group	Zone 1	II B; T3	
5	Ambient temp (Min/Max)	Relative Humidity	21.4°C to 33.5°C	100 %	
6	Other Conditions		Frequent rain, Harmattan		
7	Design	Self-supporting (stand-alone)		Applicable Specifications:	
8	Design Code	ASME Boiler and Pressure Vessel Code, SectionVIII, Div 1		EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-004 SPECIFICATION FOR VENT STACK	
9	**DESIGN PARAMETERS**				
10	Medium		Natural Gas		
11	Design	Temperature		**-39/110°C**	
12		Pressure		10 barg	
13		Capacity		HOLD 1	
14					
15	Operating	Temperature			
16		Pressure	Stack Tip	atmospheric	
17			Body Structure	atmospheric	
18		Capacity		43710.47 kg/h (HOLD 1)	(Note 4)
19	Hydrostatic Testing	Body of Base Structure		Per Code	
20		Vent Stack Extension		Per Code	
21	Gas Velocity at Vent Stack Tip			150 m/s	(Note 2)
22	Sound Pressure Level			85 dB(A) max	(Note 2)
23	Height of the Vent Stack			15 m (HOLD 1)	(Note 3)
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25	Vent Stack Riser Diameter			304.8 mm (12 inches) (HOLD 1)	
26	Maximum Sterile Area radius			HOLD1	
27	Max. back pressure at base of VentStack			0.07 barg	
28	Vent Stack Tip Type			Rain Deflector with Bird Screen	
29	**GAS Composition**				
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31	**Mechanical Design Data**				
32	Test Pressure (barg)	Internal Vacuum (mm Hg)	Per UG 99b ASME Code	***N/A***	
33	Internal Diameter of Shell (mm)	Height of Vent Stack	1016 mm (40 inches)	15000	
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36	Corrosion Allowance (mm)	Joint Efficiency	3	1	
37	Head Form	Seismic Design	2:1 Ellipsoidal	None	
38	Vessel Orientation	Total Volume Max. Oper. (m^3)	Vertical	N/A	
39	Empty Weight (kg)	Weight Full of Water (kg)	3600 (VTC)	8000 (VTC)	
40	Internals	Explosion Blast Load (kN/m^2)	N/A	None	
41	Insulation Thickness (mm)	Design Wind Speed (m/s)	NA	40	
42	**Material Specification (Note 4)**				
43	Body Structure	Head	SA 516 Gr 70 (S5)	SA 516 Gr. 70 (S5)	
44	Vent Stack Riser/Tip	Concentric Reducer	SA 333 Gr6	SA 420 WPL6	
45	Lining / Cladding	Skirt	N/A	SA 516 Gr. 70 (S5)	
46	Structural Shapes	Internals	SA 516 Gr 70 (S5)	N/A	
47	Gaskets	Lifting Lugs	Spiral Wound 316 SS	SA 516 Gr 70 (S5)	
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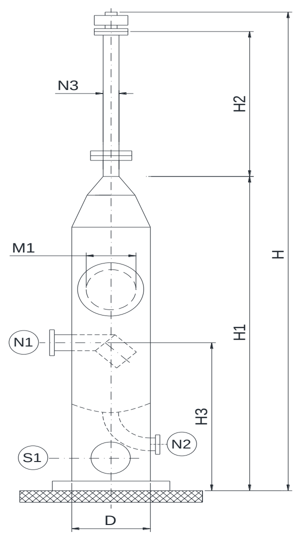
49	External Bolts	External Nuts	SA 320 L7	SA 194 7 (S3)	
50	Internal Bolts	Internal Nuts	NA	NA	
51	Plates & Pads	Drip Tray	SA 516 Gr 70 (S5)	N/A	
52	Vortex Breaker	Inlet Deflector	N/A	N/A	
53	**Fabrication and Inspection Requirements**				
54	Construction in accordance with:	Post Weld Heat Treatment	ASME Sect VIII, Div 1	As per code	
55	Inspection	Radiography	Per Code Requirement	100%	
56	Inspection Authority	Material Certificates	TPI	Per EN 10204 (Note 7)	
57	Special Heat Treatment	Ultrasonic	VTA	100%	
58	Magnetic Particle	Impact Tests	100%	Yes (Note 8)	
59	Dye Penetrant	Hydrostatic Testing	100%	Per Code Requirement	
60	**Other Requirements**				
61	Anchor Bolts	PWHT Required	ASTM F1554 (Note 15)	Per Code	
62	Fireproofing Support	Nameplate	Note 12	316 SS	
63	Drip Tray	Painting	N/A	Note 17	
64	Earthing Bosses/Clips	Lifting Lugs	Note 10	Note 11	
65	Ladder, Platform, Piping Clips:	Stiffening Rings	N/A	N/A	

MECHANICAL DATA SHEET FOR VENT STACK-AKABEL

66 **SKETCH**



VENT STACK DIMENSIONS

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	Notes
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92	1) Vendor's scope shall include complete equipment design, material, fabrication,inspection and testing.
93	2) At the tip of the Vent Stack.
94	3) From bottom surface of the skirt's baseplate up to the tip of the vent stack extension.
95	4) At full blowdown load.
96	5) Flanges shall be delivered in accordance standard ASME B16.5, RF, WN, Class 150
97	6) Material for flange: ASTM A350-LF2.
98	7) The material test certificates shall be in accordance with EN10204 Type 3.1 for all metals welded to the pressure part.
99	8) Impact test shall be made in accordance with ASTM A350 and ASTM A333, test temperature shall be -45°C.
100	9) Vent Stack shall be connected to the station's grounding system
101	10) Two earthing clips shall be provided at the skirt, approximately 500 mm above the bottom surface of the skirt's foundation flange
102	***11) Separate lifting and tailing lugs/trunions shall be provided for the top portion.***
103	***12) deleted***
104	13) All nozzles with size 2"and below shall be provided with 2 stiffeners 90 deg apart welded to nozzle neck and shell.
105	14) Any Support Cleat or Pad directly welded with vessel shall be same or equivalent material as vessel.
106	15) Anchor bolt design shall be carried out by vendor and relevant documentations to aid foundation design provided.
107	16) Hydrostatic test requirements shall be as per UG 99b of ASME Section VIII, Division 1 Code.
108	17) Painting shall comply with the requirements of the Painting and Coating Specification referenced herein.
109	18.) Vent Stack extension shall be equipped with vibration damper to reduce vibration at the tip of the stack.
110	***19) Vent stack shall be U stamped in accordance with ASME Section VIII Div.1.***
111	***20) Requirement of lightening arrestor to prevent auto ignition during lightening and emergency depressurization (considering API 520 and NFPA 780 requirements) shall be provided by Vendor. Vendor to also review the probability of auto ignition due to static charge built up and recommend/provide preventive measure.***
112	***21) Design of vent stack tip shall be carried out by vendor for properly dilution of gases in atmosphere at below lower flammable limits.***
113	***22) Vendor to carry out vibration analysis and limit the wind deflection. Reducer and flaring of skirt shall be provided to limit the wind deflection by H/200m where H is overall height of stack from base.***
114	***23) Vendor to provide saddle support for transportation of Cold Vent stack.***
115	***24) Supply of blinds required for hydrotest shall be considered and supplied by Vendor.***
116	***25) Vendor shall evaluate the need of a strengthening arrangement (guy wires, stiffeners etc.) due to high winds and the resultant stresses on such slender structure. If required, guy wires along with their accessories shall be supplied by the vent stack Vendor.***
117	***26). Vent Stack shall be equipped with Spark Arrestor and Bird Screen. Vendor to provide.***
118	**References**
119	1. EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 Basis for Design
120	2. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-004 Specification for Vent Stack
121	3. EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 Process Flow Sheets 1 to 8
122	4. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005 Piping Class Specification
123	**Abbreviations**
124	VTA - Vendor To Advise, VTC, Vendor to Confirm, N/A - Not Applicable